

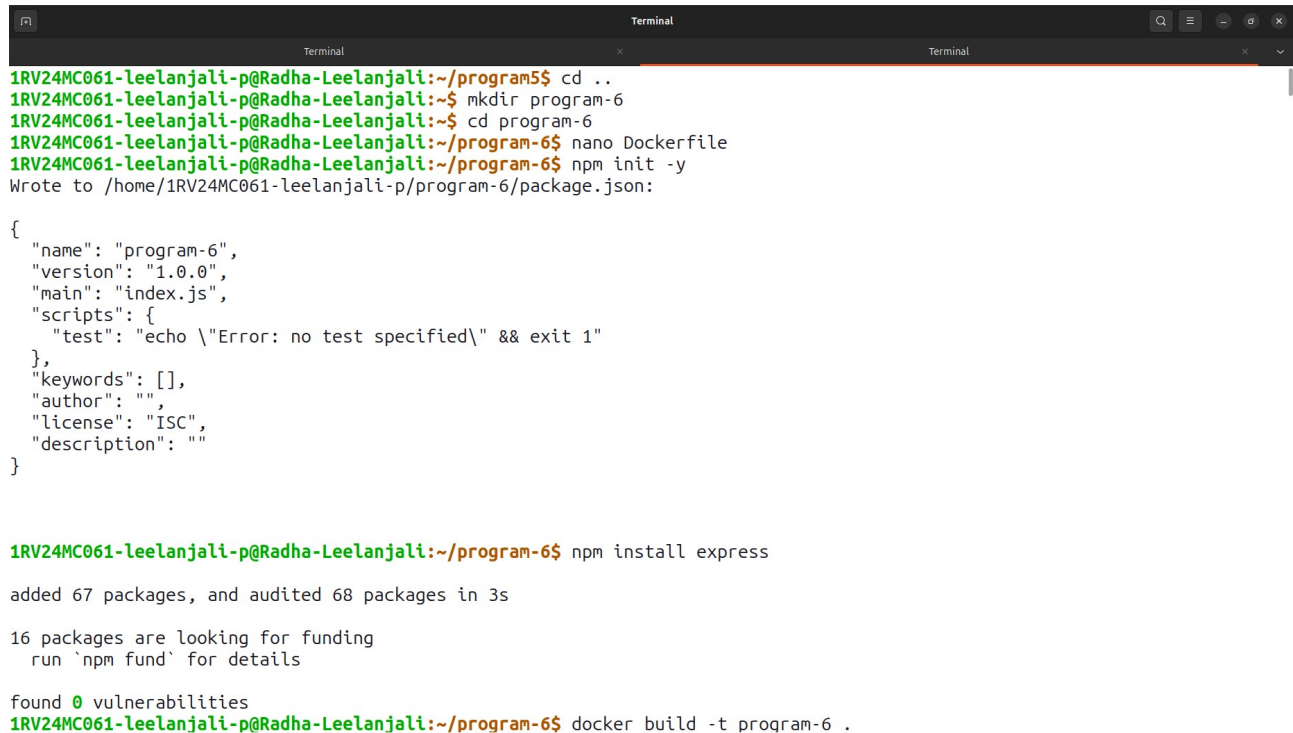
## Program-6 Documentation

**Title:** Deploy an automated build pipeline using Docker Hub

This document outlines the step-by-step procedure for creating an application, containerizing it using Docker, and pushing the resulting image to both Docker Hub and the GitHub Container Registry (GHCR), establishing a foundational CI/CD pipeline.

### 1. Local Project Preparation (program-6)

The first stage involves setting up the local directory and initializing the application code.



```
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/program5$ cd ..
1RV24MC061-leelanjali-p@Radha-Leelanjali:~$ mkdir program-6
1RV24MC061-leelanjali-p@Radha-Leelanjali:~$ cd program-6
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/program-6$ nano Dockerfile
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/program-6$ npm init -y
Wrote to /home/1RV24MC061-leelanjali-p/program-6/package.json:

{
  "name": "program-6",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}

1RV24MC061-leelanjali-p@Radha-Leelanjali:~/program-6$ npm install express

added 67 packages, and audited 68 packages in 3s

16 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/program-6$ docker build -t program-6 .
```

### 2. Source Code Management (Git & GitHub)

The project code is committed to a Git repository and pushed to GitHub for version control and to prepare for potential automated builds (like the one shown in ).

```
Terminal
Terminal
Terminal
Password:
Login Succeeded
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/program-6$ docker push leelanjai/program-6:latest
The push refers to repository [docker.io/leelanjai/program-6]
f9dbefed1f1: Pushed
be7c0b85f782: Pushed
436272e52630: Pushed
d432cd0632c7: Layer already exists
8bc61164599f: Layer already exists
a81608eb20af: Layer already exists
1548c3f692a1: Layer already exists
256f393e029f: Layer already exists
latest: digest: sha256:1fbd36fd083f73d579e6b39489e5f9287d563197917ed3e9363e7c04a6ca26a2 size: 1993
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/program-6$ docker tag program-6 ghcr.io/rvcestudent/program-6:latest
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/program-6$ echo ghp_2Vf4d2p3bLSYjJrawrhUuuJfMW5nMc3WVpBV | docker login ghcr.io
-u /rvcestudent --password-stdin

WARNING! Your credentials are stored unencrypted in '/home/1RV24MC061-leelanjali-p/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/program-6$ docker push ghcr.io/rvcestudent/program-6:latest
The push refers to repository [ghcr.io/rvcestudent/program-6]
f9dbefed1f1: Pushed
be7c0b85f782: Pushed
436272e52630: Pushed
d432cd0632c7: Layer already exists
8bc61164599f: Layer already exists
a81608eb20af: Layer already exists
1548c3f692a1: Layer already exists
```

### 3. Docker Image Creation (Local Build)

The application and its environment (defined in the `Dockerfile`) are packaged into a single, runnable Docker image.